

STA258
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Analysis of Variance
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One Sample $\longrightarrow$

σ not known

$$t^* = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Two Sample

σ_1, σ_2 both not known

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$$z^* = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

Generalizing to 3 or more groups
changes our approach.

Short answer:

As we consider more groups,
we are more likely to encounter
differences

Therefore we adapt our
approach to account for this

ANOVA (Analysis of Variance)

used to compare the means of multiple groups
(most useful for ≥ 3 groups)

ANOVA model

$$Y_{ij} = \underbrace{\mu}_{\text{overall mean}} + \underbrace{\tau_i}_{\text{treatment effect}} + \epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, \sigma^2)$$

or

$$Y_{ij} = \underbrace{\mu_i}_{\text{group mean}} + \epsilon_{ij}$$

tested in
STA 305

$j \geq 2$ obs # within a group

i # groups

	group 1	group 2	- - - - -	group k
	x_{11}	x_{21}		x_{k1}
	x_{12}	x_{22}		x_{k2}
	$\vdots$	$\vdots$		$\vdots$
	x_{1n_1}	x_{2n_2}		x_{kn_k}
mean	$\bar{x}_1$	$\bar{x}_2$		$\bar{x}_k$
st. dev	s_1	s_2		s_k
size	n_1	n_2		n_k

total sample size $= n_1 + n_2 + \dots + n_k = n$

Overall sample mean $\bar{x} = \frac{1}{n} \sum_{i=1}^k n_i \bar{x}_i$

Variation within groups

Sum Sq error

$$SSE = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2 \quad (SSW)$$

$$= \sum_{i=1}^k (n_i - 1) \cdot s_i^2$$

Variation between groups (treatment)

Sum Sq treatment

$$SSTrt = \sum_{i=1}^k (n_i \bar{x}_i - \bar{x})^2$$

Total variability

$$SSTotal = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x})^2$$

$$SSTotal = SSE + SSTrt$$

Use SSE, SSTrt, SSTotal to
complete an ANOVA table

ANOVA Table

Source	degrees of freedom	Sum of Squares	Mean Square	F-Stat
Treatment	# groups - 1 $k - 1$	SS_{Tot}	$MS_{Tot} = \frac{SS_{Tot}}{k - 1}$	$F^* = \frac{MS_{Tot}}{MSE}$
Error	# total sample - # groups $n - k$	SSE	$MSE = \frac{SSE}{n - k}$	X
Total	$n - 1$	SS_{Total}	X	X



Hyp test with ANOVA

$$H_0: \mu_1 = \mu_2 = \dots = \mu_k$$

all group means
are equal

$$H_a: \exists i, j \text{ s.t. } \mu_i \neq \mu_j$$

$$i = 1, \dots, k$$

$$j = 1, \dots, k$$

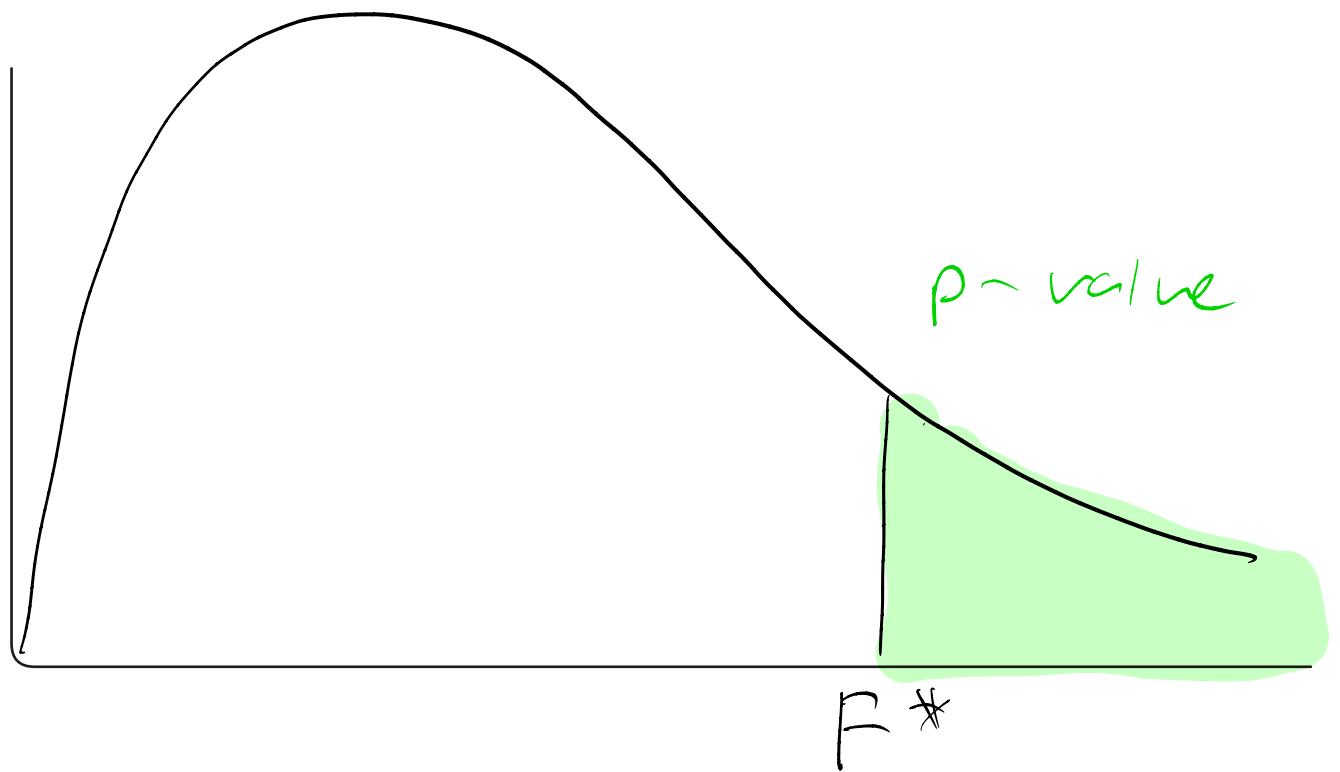
$$i \neq j$$

at least one
mean is
different

Test Stat

$$F^* = \frac{MS_{Tot}}{MSE} = \frac{SS_{Tot} / (k-1)}{SSE / (n-k)} \quad \begin{matrix} \text{num} \\ \text{denom} \end{matrix}$$

$$F^* \sim F_{\substack{k-1 \\ \text{num}}, \substack{n-k \\ \text{denom}}}$$



Conclusion! usual way

First stage analysis

$$H_0: \mu_1 = \mu_2 = \dots = \mu_k$$

FTB $H_0 \rightarrow$ Stop

If we reject H_0 , we know at least one mean is different. which mean?

$\rightarrow$ conduct post-hoc pair wise comparisons, (i.e. compare all pairs of means)

Illustrative Ex

3 groups

$$H_0: \mu_1 = \mu_2 = \mu_3$$

$$H_a: \mu_1 = \mu_2 \neq \mu_3$$

or

$$\mu_1 \neq \mu_2 = \mu_3$$

or

$$\mu_1 = \mu_3 \neq \mu_2$$

or

$$\mu_1 \neq \mu_2 \neq \mu_2$$

The Data Matrix

The following table shows last year's sales data for a small business. The sample is put into a matrix format in which each of the three columns corresponds to one of the three countries in which the company does business. The numbers in the cell represent the sales (in units of \$1000) made in that country last year. These data will be used to develop the theory underlying **Analysis of Variance** or, for short, **ANOVA**.

The Data Matrix

	Country A	Country B	Country C
	6	10	14
	10	8	13
	7	12	11
	9	10	10
Average	8	10	12

Altogether, there were 12 sales last year that totaled \$120 - so the average sale was \$10.

The column (country) averages are:

Column 1 (Country A) \$8.

Column 2 (Country B) \$10.

Column 3 (Country C) \$12.

Now we will begin our study of how to make a statistically valid prediction of the next sales figure. In that regard, there are two possible situations that can occur.

1. The country of the next sale (observation) is **not** known.
2. The country of the next sale is known.

Situation 1.

Without any additional information, the best prediction is the sample mean \$10. This prediction is best in the least squares sense - that is, if \$10 had been used to predict each of the 12 observations in the sample, then the total of the squared errors SS_{TOTAL} would be as small as possible. In our data set, SS_{TOTAL} equals 60. That figure can be verified by calculating $\sum (x_i - 10)^2$ for each observation x_i of the sample.

Situation 2. One-factor ANOVA Model.

If the country of the next sale is known, then two different predictions are possible for the next sales figure:

- The sample mean \$10.
- The mean of the sales of the country in which the next sale will occur. (In this case, \$8 if the next sale will occur in Country A, \$10 in Country B, or \$12 in Country C.) This prediction **ignores** the information present in the sales figures from the other two countries.

The Null Hypothesis for One-Factor ANOVA

We have discussed the prediction possibilities for one-factor ANOVA models. Now, we will learn how to test the statistical significance of a one-factor ANOVA model.

Let's suppose that we want to predict the next sales figure, and that we know the country in which this sale will occur. Without any statistical testing, we can always by default use the sample mean \$10 to predict the next sale. The default prediction, the sample mean, doesn't use any information about the country (column) in which the sale will occur.

The Null Hypothesis for One-Factor ANOVA

However, if instead we use the mean of the observations in only one column (the column that corresponds to the particular country in which we know the next sale will occur), then we have to test the null hypothesis

$$H_0 : \mu_{COL1}, \mu_{COL2}, \mu_{COL3}, \text{ are equal}$$

and reject it in favor of the alternative hypothesis

$$H_a : \mu_{COL1}, \mu_{COL2}, \mu_{COL3}, \text{ are NOT all equal}$$

The Null Hypothesis for One-Factor ANOVA

If the null hypothesis is rejected, then we can be statistically confident that the column means are not all equal, and therefore that the individual column means (i. e., \$8, \$10, \$12) can be used to predict the amount of the next sale. If the next sale was going to occur in Country A, then the prediction would be \$8. If the next sale was going to occur in Country B, then the prediction would be \$10. If the next sale was going to occur in Country C, then the prediction would be \$12.

The One-Factor ANOVA F Test

To test the null hypothesis stated above, we have to calculate an F-statistic. If $F_* > F_{(c-1, n-c), \alpha}$, then reject H_0 , and use the sample column means to predict future observations. Otherwise, do not reject H_0 and use the overall sample mean to predict future observations.

ANOVA Table

To see how this F_* is calculated, see the ANOVA Table below.

Source of Variation	Degrees of Freedom (df)	Sum of Square (SS)	Mean Sum of Squares (MSS)	F Ratio
Treatments	$c - 1$	SST	$\frac{SST}{c-1}$	$F = \frac{MST}{MSE}$
Error	$n - c$	SSE	$\frac{SSE}{n-c}$	
Total	$n - 1$	SS_{TOTAL}		

Calculation of SS_{TOTAL}

If **no** model is used, then the predictions for each of the 12 observations will be 10. If these predictions are used, the squared error of these 12 predictions is given in the table below.

Country A	Country B	Country C
16	0	16
0	4	9
9	4	1
1	0	0

Prediction Errors Squared when NO Factor is used (Total) = 60.

Calculation of SSE

If the column model is used, then the 12 observations would have the following 12 predictions, where \$8 is the average for the first column, \$10 is the average for the second column, and \$12 is the average for the third column.

Country A	Country B	Country C
8	10	12
8	10	12
8	10	12
8	10	12

Calculation of SSE

Using the above 12 predictions, the errors squared are shown in the table below.

Country A	Country B	Country C
4	0	4
4	4	1
1	4	1
1	0	4

Errors Squared when the Column Factor is used (Total) = 28.

Calculation of SST

The units explained by the column model are calculated by finding the square of each prediction change when moving from NO model to the column model. The following table presents the square of each prediction change:

Country A	Country B	Country C
4	0	4
4	0	4
4	0	4
4	0	4

Table of the Square of the Prediction Change when Moving from NO Model to the Column Model (Total) = 32.

ANOVA Table

The ANOVA Table for the column factor can now be filled in as shown below:

Source of Variation	Degrees of Freedom (df)	Sum of Square (SS)	Mean Sum of Squares (MSS)	F Ratio
Treatments	2	32	$\frac{32}{2} = 16$	$\frac{16}{3.1111} = 5.1428$
Error	9	28	$\frac{28}{9} = 3.1111$	
Total	11	60		

So for this one-factor ANOVA model, $F_* = 5.1428$.

Conclusion

If the null hypothesis is true, then the F-statistic should be a value from the $F_{2, 9}$ distribution. Referring to the table that contains the upper 0.05 cut-off points of F distributions, we see that $F_{(2,9),0.05} = 4.256$. Since 5.1428 is greater than 4.256, this tells us that the F-statistic is in the upper 0.05 of the $F_{2, 9}$ distribution. Therefore we reject the null hypothesis at the 0.05 significance level, and we conclude that the country means are not all the same. Thus, the prediction for the next sale in a known country is the mean of all the previous sales in that country.

R Code;

```
salesA=c(6,10,7,9);
```

```
salesB=c (10, 8,12,10);
```

```
salesC=c(14,13,11,10);
```

```
sales=c(salesA,salesB,salesC);
```

```
country=c(rep(1,4),rep(2,4),rep(3,4));
```

```
oneway.test(sales~country,var.equal=TRUE);
```



```
##  
## One-way analysis of means  
##  
## data: sales and country  
## F = 5.1429, num df = 2, denom df = 9, p-value =  
## 0.0324
```

```
# R Code;  
# Another way: using lm;  
  
salesA=c(6,10,7,9);  
  
salesB=c (10, 8,12,10);  
  
salesC=c(14,13,11,10);  
  
sales=c(salesA,salesB,salesC);  
  
country=c(rep(1,4),rep(2,4),rep(3,4));  
  
anova(lm(sales~factor(country)));
```

```
## Analysis of Variance Table
##
## Response: sales
##              Df Sum Sq Mean Sq F value Pr(>F)
## factor(country)  2      32 16.0000  5.1429 0.0324 *
## Residuals       9      28  3.1111
## ---
## Signif. codes:
## 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Underlying Assumptions

Officially, to use the predictions from an ANOVA model, three assumptions about the populations from which the sample was taken must be satisfied:

1. Each population has a Normal distribution.
2. Each population has the **same** standard deviation σ .
3. The observations are mutually independent of one another.

Sum of Squares for Treatments (a.k.a. between-treatments variation or Explained)

$$SST = \sum_{j=1}^k n_j (\bar{x}_j - \bar{\bar{x}})^2$$

Sum of Squares for Error (a.k.a. within-treatments variation or Unexplained)

$$SSE = \sum_{j=1}^k \sum_{i=1}^{n_j} (x_{ij} - \bar{x}_j)^2 = (n_1 - 1)s_1^2 + \dots + (n_k - 1)s_k^2.$$

Mean Square for Treatments

$$MST = \frac{SST}{k - 1}$$

Mean Square for Error

$$MSE = \frac{SSE}{n - k}$$

Test Statistic

$$F = \frac{MST}{MSE}$$

Exercise

A statistics practitioner calculated the following statistics:

Statistic	Treatment		
	1	2	3
n	5	5	5
$\bar{x}$	10	15	20
s^2	50	50	50

Complete the ANOVA table.

Solution

$$\bar{\bar{X}} = \frac{5(10)+5(15)+5(20)}{5+5+5} = 15$$

$$SST = 5(10 - 15)^2 + 5(15 - 15)^2 + 5(20 - 15)^2 = 250$$

$$SSE = (5 - 1)(50) + (5 - 1)(50) + (5 - 1)(50) = 600$$

ANOVA Table

Source of Variation	Degrees of Freedom (df)	Sum of Square (SS)	Mean Sum of Squares (MSS)	F Ratio
Treatments	2	250	$\frac{250}{2} = 125$	$\frac{125}{50} = 2.50$
Error	12	600	$\frac{600}{12} = 50$	
Total	14	850		

Exercise

A statistics practitioner calculated the following statistics:

Statistic	Treatment		
	1	2	3
n	4	4	4
$\bar{x}$	20	22	25
s^2	10	10	10

Complete the ANOVA table.

Solution

$$\bar{\bar{X}} = \frac{4(20) + 4(22) + 4(25)}{4 + 4 + 4} = 22.33$$

$$SST = 4(20 - 22.33)^2 + 4(22 - 22.33)^2 + 4(25 - 22.33)^2 = 50.67$$

$$SSE = (4 - 1)(10) + (4 - 1)(10) + (4 - 1)(10) = 90$$

ANOVA Table

Source of Variation	Degrees of Freedom (df)	Sum of Square (SS)	Mean Sum of Squares (MSS)	F Ratio
Treatments	2	50.67	$\frac{50.67}{2} = 25.33$	$\frac{25.33}{10} = 2.53$
Error	9	90	$\frac{90}{9} = 10$	
Total	11	140.67		

Exercise

A consumer organization was concerned about the differences between the advertised sizes of containers and the actual amount of product. In a preliminary study, six packages of three different brands of margarine that are supposed to contain 500ml were measured. The differences from 500 ml are listed here. Do these data provide sufficient evidence to conclude that differences exist between the three brands? Use $\alpha = 0.05$.

Brand 1	Brand 2	Brand 3
1	2	1
3	2	2
3	4	4
0	3	2
1	0	3
0	4	4

balanced

$$k=3$$

Brand 1	Brand 2	Brand 3
1	1	1
3	1	1
3	1	1
1	1	1
1	1	1

$$d=0.05$$

$$n_1=6$$

$$n_2=6$$

$$n_3=6$$

$$\bar{x}_1=1.33$$

$$\bar{x}_2=2.50$$

$$\bar{x}_3=2.67$$

$$s_1^2=1.87$$

$$s_2^2=2.30$$

$$s_3^2=1.47$$

Total sample size

$$n=n_1+n_2+n_3=18$$

Find overall mean

$$\bar{x} = \frac{1}{n} \sum n_i \bar{x}_i = \frac{1}{18} [(6)(1.33) + (6)(2.50) + (6)(2.67)]$$

$$= 2.17$$

$$SSE = \sum_{i=1}^k (n_i - 1) s_i^2 = (6-1)(1.87) + (6-1)(2.30) + (6-1)(1.47) \\ = 28.20$$

$$SST_{\text{trt}} = \sum n_i (\bar{x}_i - \bar{x})^2 \\ = 6(1.33 - 2.17)^2 + 6(2.50 - 2.17)^2 + 6(2.67 - 2.17)^2 \\ = 6.39$$

not necessary to calculate SS_{total}

Source	df	SS	MS	F
Treatment	$k-1$ $= 3-1$ $= 2$	SS_{Trt} $= 6.39$	MS_{Trt} $= \frac{6.39}{2}$ $= 3.195$	$F^* = \frac{3.195}{1.88}$ $= 1.70$
Error	$n-k$ $= 18-3$ $= 15$	SSE $= 28.20$	MSE $= \frac{28.20}{15}$ $= 1.88$	X
Total	17	$SS_{Tot} = 34.59$	X	X

$$H_0: \mu_1 = \mu_2 = \mu_3$$

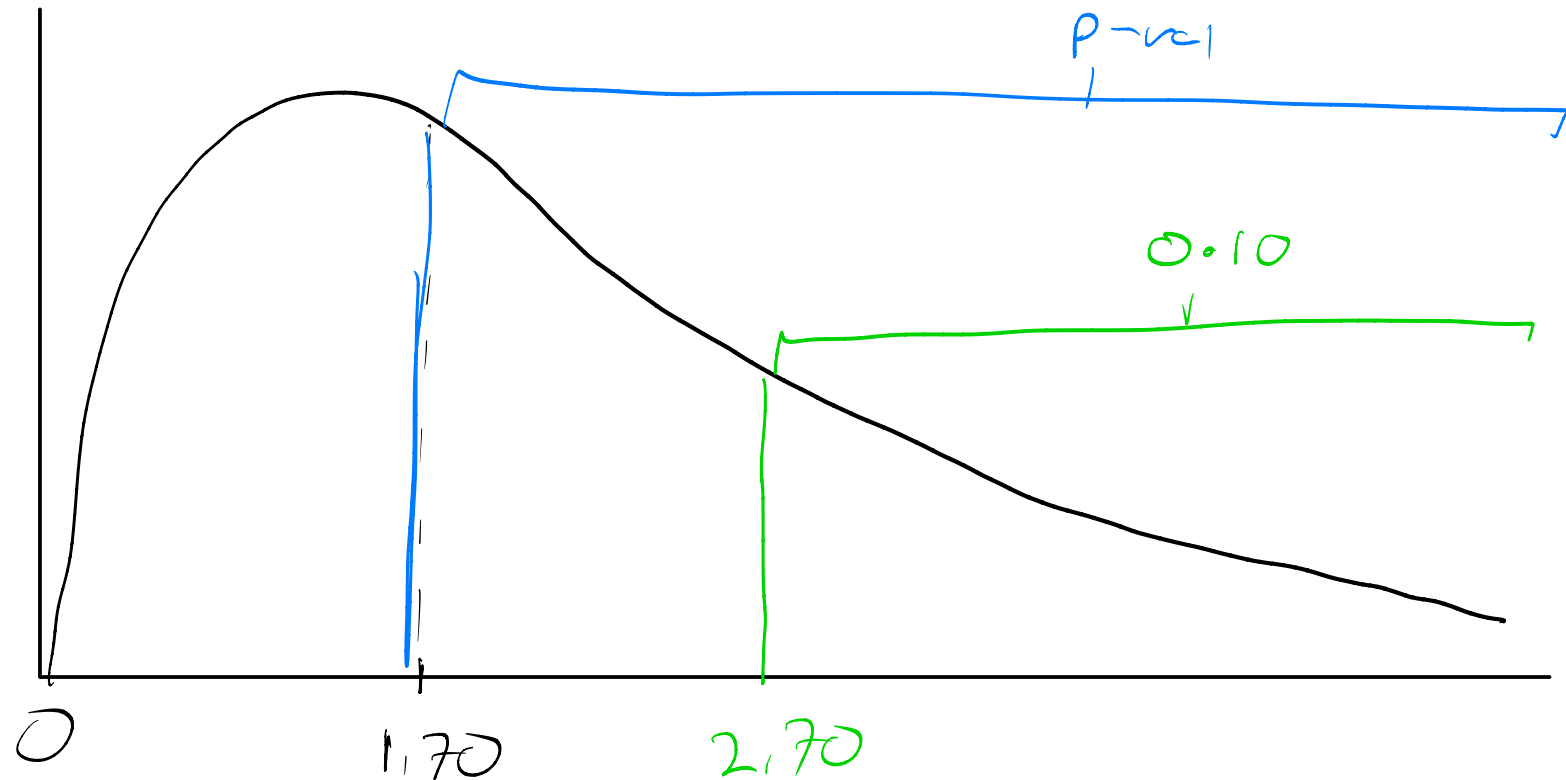
H_a : at least one mean is different

$$F = \frac{MST}{MSE} = \frac{SST / (k-1)}{SSE / (n-k)}$$

Test stat

$$F^* = 1.70 \sim F_{\substack{2, 15 \\ k-1, n-k}}$$

Using F at num = 2 df and denom = 15 df



$$p\text{-val} > 0.10 > 0.05$$

(d)

Do not reject H_0 . Insufficient evidence to suggest any group means are different.

STOP

Assumptions

1/ Indep. Samples

2/ Random Samples

3/ Assume all groups have the same variance

$$\sigma_1^2 = \sigma_2^2 = \dots = \sigma_k^2 = \sigma^2$$

Estimate common variance using

$$\hat{\sigma}^2 = \text{MSE}$$

from ANOVA

Consider comparing 3 groups

μ_1, μ_2, μ_3

reject $H_0: \mu_1 = \mu_2 = \mu_3$

Which μ 's are different?

compare $\mu_1 - \mu_2$
 $\mu_1 - \mu_3$
 $\mu_2 - \mu_3$

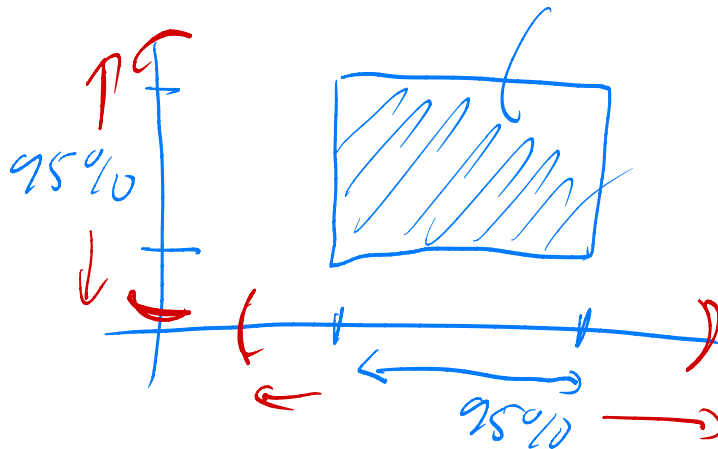
$$\binom{3}{2} = 3 \text{ comparisons}$$

create simultaneous $C\%$ CI's for each pair of difference.

suppose we want this system to have 95% confidence.

consider 2 intervals

$$(0.95)(0.95) = 0.9025 < 0.95$$



Perform Bonferroni Correction

Similar to pooled CI

ANOVA

$$\sigma^2 = \text{MSE}$$

$$\sigma = \sqrt{\text{MSE}}$$

pooled CI

$$(\bar{x}_1 - \bar{x}_2) \pm t_{(n_1+n_2-2, \alpha/2)} \cdot s_p \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

With Bonferroni

$$(\bar{x}_i - \bar{x}_j) \pm t_{(n-k, \alpha/2m)} \cdot \sqrt{\text{MSE}} \cdot \sqrt{\frac{1}{n_i} + \frac{1}{n_j}}$$

$$m = \binom{k}{2}$$

pairwise comparisons

Step 1. State Hypotheses.

μ_i = population mean for differences from 500 ml (brand i , where $i = 1, 2, 3$).

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

H_a : At least two means differ.

Step 2. Compute test statistic.

	Brand 1	Brand 2	Brand 3
Mean	1.33	2.50	2.67
Variance	1.87	2.30	1.47

Grand mean = $\bar{\bar{x}} = 2.17$.

$$SST = 6(1.33 - 2.17)^2 + 6(2.50 - 2.17)^2 + 6(2.67 - 2.17)^2 = 6.387 \approx 6.39$$

$$SSE = (6 - 1)(1.87) + (6 - 1)(2.30) + (6 - 1)(1.47) = 28.20$$

Grand mean = $\bar{\bar{x}} = 2.17$.

$$SST = 6(1.33 - 2.17)^2 + 6(2.50 - 2.17)^2 + 6(2.67 - 2.17)^2 = 6.387 \approx 6.39$$

$$SSE = (6 - 1)(1.87) + (6 - 1)(2.30) + (6 - 1)(1.47) = 28.20$$

ANOVA Table

Source of Variation	Degrees of Freedom (df)	Sum of Square (SS)	Mean Sum of Squares (MSS)	F Ratio
Treatments	2	6.39	$\frac{6.39}{2} = 3.195$	$\frac{3.195}{1.88} = 1.70$
Error	15	28.20	$\frac{28.20}{15} = 1.88$	
Total	17	34.59		

Step 3. Find Rejection Region.

We reject the null hypothesis only if

$$F > F_{\alpha, k-1, n-k}$$

If we let $\alpha = 0.05$, the rejection region for this exercise is

$$F > F_{0.05, 2, 15} = 3.682$$

Step 4. Conclusion.

We found the value of the test statistic to be $F = 1.70$. Since $F = 1.70 < F_{0.05, 2, 15} = 3.682$, we **can't** reject H_0 . Thus, there is **not** evidence to infer that the average differences differ between the three brands.

```
# R Code;
```

```
brand1=c(1,3,3,0,1,0);
```

```
brand2=c (2,2,4,3,0,4);
```

```
brand3=c(1,2,4,2,3,4);
```

```
differences=c(brand1,brand2,brand3);
```

```
brand=c(rep(1,6),rep(2,6),rep(3,6));
```

```
oneway.test(differences~brand,var.equal=TRUE);
```

```
##  
## One-way analysis of means  
##  
## data: differences and brand  
## F = 1.6864, num df = 2, denom df = 15, p-value  
## = 0.2185
```

```
> brand = as.factor(brand)
> anova_model = aov(differences~brand)
> anova(anova_model)
```

Analysis of Variance Table

Response: differences

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
brand	2	6.3333	3.1667	1.6864	0.2185
Residuals	15	28.1667	1.8778		

The friendly folks at the Internal Revenue Service (IRS) in the United States and Canada Revenue Agency (CRA) are always looking for ways to improve the wording and format of its tax return forms. Three new forms have been developed recently. To determine which, if any, are superior to the current form, 120 individuals were asked to participate in an experiment. Each of the three new forms and the currently used form were filled out by 30 different people. The amount of time (in minutes) taken by each person to complete the task was recorded.

What conclusions can be drawn from these data?

```
#Step 1. Entering data;  
# importing data;  
# url of tax return forms;  
forms_url =  
"https://mcs.utm.utoronto.ca/~nosedal/data/tax-forms.txt"  
forms_data= read.table(forms_url,header=TRUE);  
names(forms_data);  
forms_data[1:4, ];
```

```
## [1] "Form1" "Form2" "Form3" "Form4"  
##   Form1 Form2 Form3 Form4  
## 1    23    88   116   103  
## 2    59   114   123   122  
## 3    68    81    64   105  
## 4   122    41   136    73
```



```
#Step 2. ANOVA;
```

```
time1=forms_data$Form1;
```

```
time2=forms_data$Form2;
```

```
time3=forms_data$Form3;
```

```
time4=forms_data$Form4;
```

```
length(forms_data$Form1);
```

```
times=c(time1,time2,time3,time4);
```

```
forms=c(rep(1,30),rep(2,30),rep(3,30),rep(4,30));
```

```
oneway.test(times~forms,var.equal=TRUE)
```

```
## [1] 30
##
## One-way analysis of means
##
## data:  times and forms
## F = 2.9358, num df = 3, denom df = 116, p-value
## = 0.03632
```

Multiple Comparisons

Example

Because of foreign competition, North American automobile manufacturers have become more concerned with quality. One aspect of quality is the cost of repairing damage caused by accidents. A manufacturer is considering several new types of bumpers. To test how well they react to low-speed collisions, 10 bumpers of each of four different types were installed on mid-size cars, which were then driven into a wall at 5 miles per hour. The cost of repairing the damage in each case was assessed. The data are shown below.

- Is there sufficient evidence at the 5% significance level to infer that the bumpers differ in their reactions to low-speed collisions?
- If differences exist, which bumpers differ? Apply Fisher's LSD method with the Bonferroni adjustment.

Data

① Bumper 1	② Bumper 2	Bumper 3	Bumper 4
610	404	599	272
354	663	426	405
234	521	429	197
399	518	621	363
$n_1 = 10$ 278	$n_2 = 10$ 499	426	297
358	374	414	538
379	562	332	181
548	505	460	318
196	375	494	412
444	438	637	499

$$\bar{x}_1 = 380$$

$$\bar{x}_2 = 485.9$$

$$n = 40$$

```
cost_bumper1 = c(610, 354, 234, 399, 278, 358, 379, 548, 196, 444)
cost_bumper2 = c(404, 663, 521, 518, 499, 374, 562, 505, 375, 438)
cost_bumper3 = c(599, 426, 429, 621, 426, 414, 332, 460, 494, 637)
cost_bumper4 = c(272, 405, 197, 363, 297, 538, 181, 318, 412, 499)
```

```
bumper_data = rbind(
  data.frame(cost = cost_bumper1, type = "Bumper 1"),
  data.frame(cost = cost_bumper2, type = "Bumper 2"),
  data.frame(cost = cost_bumper3, type = "Bumper 3"),
  data.frame(cost = cost_bumper4, type = "Bumper 4")
)
```

```
# data in long form at
bumper_data
```

```
# Create anova model
```

```
bumper_model = aov(cost ~ type, data = bumper_data)
```

```
# Get ANOVA table
```

```
anova(bumper_model)
```

Analysis of Variance Table

Response: cost

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
type	3	150884	50295	4.0563	0.01395 *
Residuals	36	446368	12399		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$$k-1 = 3$$
$$k = 4$$

$$n - k = 36$$

$$n - 4 = 36$$

$$n = 40$$

SSE

MSE

$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$ H_a : at least one diff

reject H_0 . which means are different?

$$\binom{k}{2} = \binom{4}{2} = 6 \text{ comparisons}$$

$$\mu_1 - \mu_2$$

$$\mu_1 - \mu_3$$

$$\mu_1 - \mu_4$$

$$\mu_2 - \mu_3$$

$$\mu_2 - \mu_4$$

$$\mu_3 - \mu_4$$

Bonferroni for $\mu_1 - \mu_2$

95% simultaneous CI

$$(\bar{x}_1 - \bar{x}_2) \pm t_{(n-k, \alpha/2m)} \cdot \sqrt{MSE} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$m = \binom{k}{2} = \binom{4}{2} = 6$$

$$95\% \text{ CI} \rightarrow 1 - \alpha = 0.95$$

$$\alpha = 0.05$$

$$\frac{\alpha}{2m} = \frac{0.05}{(2)(6)} = 0.004167$$

$$n = 40, \quad n - k = 40 - 4 = 36$$

$$t_{(n-k, \alpha/2m)} = t_{(36, 0.004167)} = 2.792$$

obtained from R

$$(\bar{x}_1 - \bar{x}_2) \pm t_{(n-k, \alpha/2m)} \cdot \sqrt{MSE} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

from ANOVA

$$= (380 - 485.9)$$

$$\pm (2.792) (\sqrt{12399}) \sqrt{\frac{1}{10} + \frac{1}{10}}$$

$$= -105.9 \pm 139.0335$$

$$= (-244.9335, +33.13347)$$

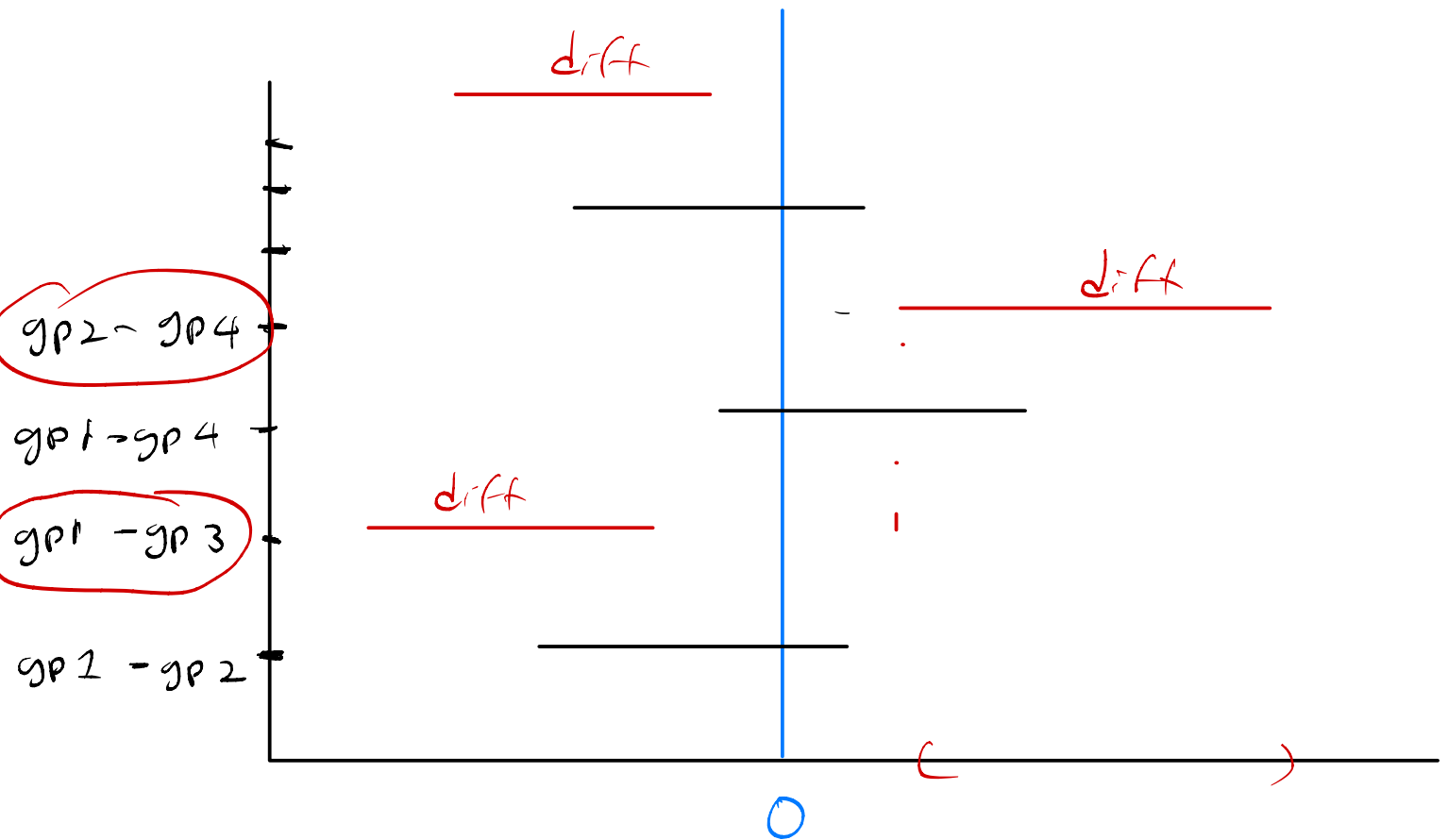
(-)

(+)

This particular Bonferroni CI

suggests μ_1 and μ_2 are not different FO inside

Visual Representation of Bonferroni



Solution a)

The test statistic is $F_* = 4.06$ and the $P - value = 0.0139$. There is enough statistical evidence to infer that there are differences between some of the bumpers. The question is now, Which bumpers differ?

Fisher's Least Significant Difference Method

The test statistic to determine whether μ_i and μ_j differ is

$$t_{ij} = \frac{(\bar{x}_i - \bar{x}_j)}{\sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}}$$

with degrees of freedom $\nu = n - k$, where n = total number of observations and k = number of samples (groups).

Fisher's Least Significant Difference Method

The confidence interval estimator is

$$(\bar{x}_i - \bar{x}_j) \pm t_{\alpha/2} \sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

Least Significant Difference (definition)

We define the least significant difference LSD as

$$LSD = t_{\alpha/2} \sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

A simple way of determining whether differences exist between each pair of population means is to compare the absolute value of the difference between their two sample means and LSD. In other words, we will conclude that μ_i and μ_j differ if

$$|\bar{x}_i - \bar{x}_j| > LSD$$

LSD will be the same for all pairs of means if all k sample sizes are equal. If some sample sizes differ, LSD must be calculated for each combination.

It can be argued that this method is flawed because it will increase the probability of committing a Type I error. That is, it is more likely that the analysis of variance to conclude that a difference exists in some of the population means when in fact none differ.

The true probability of making at least one Type I error is called the experimentwise Type I error rate, denoted α_E . The experimentwise Type I error rate can be calculated as

$$\alpha_E = 1 - (1 - \alpha)^C$$

Here C is the number of pairwise comparisons, which can be calculated by $C = \frac{k(k-1)}{2}$. It can be shown that

$$\alpha_E \leq C\alpha$$

which means that if we want the probability of making at least one Type I error to be no more than α_E , we simply specify $\alpha = \frac{\alpha_E}{C}$. The resulting procedure is called the **Bonferroni adjustment**.

Solution b)

Let's use our example to illustrate Fisher's LSD method and the Bonferroni adjustment. The four sample means and standard deviations are

$$\left\{ \begin{array}{l} \bar{y}_1 = 380 \text{ and } s_1 = 130.0931 \\ \bar{y}_2 = 485.9 \text{ and } s_2 = 90.5396 \\ \bar{y}_3 = 483.8 \text{ and } s_3 = 102.1086 \\ \bar{y}_4 = 348.2 \text{ and } s_4 = 118.5268 \end{array} \right.$$

The pairwise absolute differences are

$$|\bar{y}_1 - \bar{y}_2| = |380 - 485.9| = 105.9$$

$$|\bar{y}_1 - \bar{y}_3| = |380 - 483.8| = 103.8$$

$$|\bar{y}_1 - \bar{y}_4| = |380 - 348.2| = 31.8$$

$$|\bar{y}_2 - \bar{y}_3| = |485.9 - 483.8| = 2.1$$

$$|\bar{y}_2 - \bar{y}_4| = |485.9 - 348.2| = 137.7$$

$$|\bar{y}_3 - \bar{y}_4| = |483.8 - 348.2| = 135.6$$

Solution b)

We have that $MSE = 12,399$ and $\nu = n - k = 40 - 4 = 36$.

If we perform the LSD procedure with the Bonferroni adjustment, the number of pairwise comparisons is 6. We set

$\alpha = 0.05/6 = 0.0083$. Thus $t_{\alpha/2, n-k} = t_{0.00415, 36} = 2.7935$ (using R) and

```
qt(0.00415, 36)
```

```
## [1] -2.793555
```

$$LSD = t_{\alpha/2} \sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)} \approx 2.7935 \sqrt{12399 \left(\frac{1}{10} + \frac{1}{10} \right)} = 139.109$$

Now no pair of means differ because all the absolute values of the differences between sample means are less than 139.1095. The drawback to the LSD procedure is that we increase the probability of at least one Type I error. The Bonferroni adjustment corrects this problem.

```
library(mosaic)
```

```
result = do.call(rbind, lapply(list(cost_bumper1, cost_bumper2,  
  cost_bumper3, cost_bumper4), fav_stats))
```

```
rownames(result) = paste("Bumper", 1:4)
```

```
result
```

	min	Q1	median	Q3	max	mean	sd	n	missing
Bumper 1	196	297.00	368.5	432.75	610	380.0	130.09313	10	0
Bumper 2	374	412.50	502.0	520.25	663	485.9	90.53968	10	0
Bumper 3	332	426.00	444.5	572.75	637	483.8	102.10866	10	0
Bumper 4	181	278.25	340.5	410.25	538	348.2	118.52688	10	0

```
library(DescTools)
PostHocTest(bumper_model, method = "lsd")
```

Posthoc multiple comparisons of means : Fisher LSD
95% family-wise confidence level

\$type

			diff	lwr.ci	upr.ci	pval	
Bumper	2-Bumper	1	105.9	4.905342	206.89466	0.0404	*
Bumper	3-Bumper	1	103.8	2.805342	204.79466	0.0443	*
Bumper	4-Bumper	1	-31.8	-132.794658	69.19466	0.5271	
Bumper	3-Bumper	2	-2.1	-103.094658	98.89466	0.9666	
Bumper	4-Bumper	2	-137.7	-238.694658	-36.70534	0.0089	**
Bumper	4-Bumper	3	-135.6	-236.594658	-34.60534	0.0099	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
library(DescTools)
```

```
PostHocTest(bumper_model, method = "bonferroni")
```

Posthoc multiple comparisons of means : Bonferroni
95% family-wise confidence level

\$type

			diff	lwr.ci	upr.ci	pval
Bumper 2-Bumper 1	1	105.9	-33.13411	244.934113	0.2423	
Bumper 3-Bumper 1	1	103.8	-35.23411	242.834113	0.2657	
Bumper 4-Bumper 1	1	-31.8	-170.83411	107.234113	1.0000	
Bumper 3-Bumper 2	2	-2.1	-141.13411	136.934113	1.0000	
Bumper 4-Bumper 2	2	-137.7	-276.73411	1.334113	0.0535 .	
Bumper 4-Bumper 3	3	-135.6	-274.63411	3.434113	0.0595 .	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Example

An apple juice manufacturer has developed a new product - a liquid concentrate that, when mixed with water, produces 1 liter of apple juice. The product has several attractive features. First, it is more convenient than canned apple juice. Second, because the apple juice that is sold in cans is actually made from concentrate, the quality of the new product is at least as high as canned apple juice. Third, the cost of the new product is slightly lower than canned apple juice. The marketing manager has to decide how to market the new product. She can create advertising that emphasizes convenience, quality, or price.

Example

To facilitate a decision, she conducts an experiment. In three different cities that are similar in size and demographic makeup, she launches the product with advertising stressing the convenience of the liquid concentrate in one city. In the second city, the advertisements emphasize the quality of the product. Advertising that highlights the relatively low cost of the liquid concentrate is used in the third city. The number of packages sold weekly is recorded for the 20 weeks following the beginning of the campaign.

Example

These data are available at:

```
ad_url =  
"https://mcs.utm.utoronto.ca/~nosedal/data/ad-strategies.txt"
```

The marketing manager wants to know if differences in sales exist between the three advertising strategies.

Example

To illustrate Fisher's LSD method and the Bonferroni adjustment, consider the dataset described above and assume we tested to determine whether population means differ using a 5% significance level. The three sample means are: 577.55, 653.0, and 608.65.

The pair-wise absolute differences are

$$|\bar{x}_1 - \bar{x}_2| = |577.55 - 653.0| = |-75.45| = 75.45$$

$$|\bar{x}_1 - \bar{x}_3| = |577.55 - 608.65| = |-31.10| = 31.10$$

$$|\bar{x}_2 - \bar{x}_3| = |653.0 - 608.65| = |44.35| = 44.35$$

Example

If we conduct the LSD procedure with $\alpha = 0.05$ we find

$$|t_{\alpha/2, n-k}| = |t_{0.025, 57}| = |-2.0024655| = 2.0024655$$

```
qt(0.025, 57)
```

```
## [1] -2.002465
```

$$LSD = t_{\alpha/2} \sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)} \approx 2.002 \sqrt{8894 \left(\frac{1}{20} + \frac{1}{20} \right)} = 59.71$$

Example

We can see that only one pair of sample means differ by more than 59.71. That is, $|\bar{x}_1 - \bar{x}_2| = 75.45$, and the other two differences are less than LSD. We conclude that only μ_1 and μ_2 differ.

If we perform the LSD procedure with the Bonferroni adjustment, the number of pairwise comparisons is 3 (calculated as

$C = k(k-1)/2 = 3(2)/2$). We set $\alpha = 0.05/3 = 0.0167$. Thus, $t_{\alpha/2, n-k} = t_{0.0083, 57} = -2.4682794$ and

$$t_{\alpha/2} \sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)} \approx 2.467 \sqrt{8894 \left(\frac{1}{20} + \frac{1}{20} \right)} = 73.54$$

Again we conclude that only μ_1 and μ_2 differ. Notice, however, in the second calculation LSD is larger.

The drawback to LSD is that we increase the probability of at least one Type I error. The Bonferroni adjustment corrects this problem.